

Explicit Lattices at the Minkowski Scale with List Decoding

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Abstract. Minkowski proved that every n -dimensional lattice $\mathcal{L}$ contains a nonzero vector of Euclidean norm at most $\sqrt{n} \det(\mathcal{L})^{1/n}$. Lattices whose minimum distances approach this bound yield dense sphere packings and error-correcting codes in $\mathbb{R}^n$. Although probabilistic constructions attain the bound up to a constant factor, a deterministic polynomial-time construction in every sufficiently large dimension has remained open.

We resolve this problem by giving a deterministic polynomial-time algorithm that, for every positive integer n , constructs an integer basis of an n -dimensional lattice $\mathcal{L}$ satisfying $\lambda_1(\mathcal{L}) \geq c\sqrt{n} \det(\mathcal{L})^{1/n}$ for an absolute constant $c > 0$. We also give a deterministic list-decoding algorithm that, for every target $\mathbf{y}$, returns all lattice vectors at distance less than $\lambda_1(\mathcal{L})/\sqrt{2}$ from $\mathbf{y}$ in time polynomial in n and the encoding length of $\mathbf{y}$. Our construction combines expander graphs and shortened BCH codes to obtain binary codes of every prescribed length n and designed distance d , with codimension $O(\sqrt{nd} \log(2n/d))$ and efficient bounded-distance decoding. We arrange these codes into a nested family and apply Construction D, controlling the determinant while preserving the distance and decoding guarantees.

1 Introduction

A full rank lattice $\mathcal{L}$ in $\mathbb{R}^n$ is the set of all integer linear combinations of n linearly independent vectors. Placing a ball of radius $\lambda_1(\mathcal{L})/2$ at every point of a lattice $\mathcal{L}$ gives a sphere packing, where $\lambda_1(\mathcal{L})$ is the length of its shortest nonzero vector. Sphere packing asks how densely congruent balls can be arranged without overlap, a classical problem connecting geometry, number theory, and coding theory [5]. Major milestones include the resolution of the Kepler conjecture in dimension three [13] and the proofs of optimality of the E_8 and Leech lattice packings in dimensions 8 and 24 [4, 31].

For a full rank lattice $\mathcal{L}$, its determinant $\det(\mathcal{L})$ is the volume of a fundamental parallelepiped. The quality of its sphere packing is measured by the scale-invariant *normalized minimum distance*

$$\delta(\mathcal{L}) := \frac{\lambda_1(\mathcal{L})}{\det(\mathcal{L})^{1/n}}.$$

Minkowski's first theorem gives $\delta(\mathcal{L}) \leq \sqrt{n}$, and the Minkowski–Hlawka theorem shows that this bound is tight up to an absolute constant [5, Chapter 1].

Indeed, a random determinant-one lattice drawn from the invariant probability measure has $\lambda_1 = \Omega(\sqrt{n})$ with high probability, as follows from Siegel’s mean-value theorem [27]. Martinet’s number field lattices attain normalized minimum distance $\Omega(\sqrt{n})$ along an infinite sequence of dimensions [18]. Other algebraic constructions of dense lattices include the Mordell–Weil lattices of Shioda and Elkies [26,8]. Moustrou [22] gave an effective construction attaining Minkowski’s scale in infinitely many dimensions by finite search, with running time $\exp((3/2 + o(1))n \log n)$. These results lead to two algorithmic questions.

The first asks for a *deterministic polynomial time construction*: can one output a basis of an n -dimensional lattice with $\delta(\mathcal{L}) = \Omega(\sqrt{n})$ for every sufficiently large n ? Vaikuntanathan [30] explicitly formulated the question of polynomial time basis construction at this scale. It requires both control of the dimension and an efficient basis constructor, beyond a geometric description or an existence proof.

The second asks for *efficient decoding at Minkowski’s scale*. Given a target $\mathbf{y} = \mathbf{x} + \mathbf{e}$ with $\mathbf{x} \in \mathcal{L}$, a bounded distance decoder must recover $\mathbf{x}$ whenever $\|\mathbf{e}\|_2 < R$, for a prescribed radius $R \leq \lambda_1(\mathcal{L})/2$. Does there exist a family of lattices with a polynomial time decoder attaining $R/\det(\mathcal{L})^{1/n} = \Omega(\sqrt{n})$? The guarantee must hold for every error within the radius. Beyond the uniqueness radius $\lambda_1/2$, *list decoding* asks for all lattice points within the prescribed distance of the target.

Prior work. Efficient lattice decoding has motivated a substantial line of work. Micciancio and Nicolosi [20] gave a polynomial time decoder for Barnes–Wall lattices throughout the uniqueness radius $\lambda_1/2$. Grigorescu and Peikert [12] extended this to list decoding at radius $(1 - \tau)\lambda_1$ for every fixed $\tau \in (0, 1)$. The normalized minimum distance of these lattices is $\Theta(n^{1/4})$, so their normalized decoding radii remain below Minkowski’s scale by a polynomial factor.

Ducas and Pierrot [6] constructed discrete logarithm lattices with a polynomial time BDD algorithm achieving normalized decoding radius $\Theta(\sqrt{n}/\log n)$.

Barnes and Sloane [1] used Construction D with BCH codes to obtain normalized minimum distance $\Omega(\sqrt{n/\log n})$. Mook and Peikert [21] supplied an efficient list decoder for these lattices, combining Euclidean list decoding of BCH codes with recursive decoding of Construction D. For every fixed $\tau \in (0, 1)$, their decoder reaches $(1 - \tau)\lambda_1/\sqrt{2}$. Bennett and Peikert [2] obtained the same order of normalized decoding radius using Construction A and Reed–Solomon codes.

Kirshanova and Malygina [16] replaced the BCH tower by subfield subcodes of Garcia–Stichtenoth algebraic geometry codes, improving the normalized minimum distance to $\Omega\left(\frac{\sqrt{n}}{(\log n)^\varepsilon \sqrt{\log \log n}}\right)$ for any fixed $\varepsilon > 0$. They retain list decoding to $(1 - \tau)\lambda_1/\sqrt{2}$ for every fixed $\tau \in (0, 1)$. For each fixed ε , this guarantee still loses a superconstant factor from Minkowski’s scale. We summarize these results in Table 1.

For Gaussian noise, polar lattices [32] admit polynomial time decoding under Gaussian noise with typical Euclidean norm $\Theta(\sqrt{n} \det(\mathcal{L})^{1/n})$, with success

probability $1 - o(1)$ as $n \rightarrow \infty$. This guarantee is over the noise distribution and does not ensure successful decoding for every error vector of comparable norm.

Efficient lattice decoding also has cryptographic applications. Ducas and van Woerden [7] use efficiently decodable lattices in key encapsulation mechanisms based on the *Lattice Isomorphism Problem*. Their most attractive parameters require both the lattice and its dual to have normalized minimum distance near Minkowski's bound. Our construction provides the primal distance and decoding guarantees, but does not meet the additional requirement on the dual.

1.1 Our results

We resolve both questions by giving a deterministic polynomial time construction in every dimension that attains Minkowski's scale and admits efficient list decoding to the radius $\lambda_1/\sqrt{2}$. In particular, the resulting lattices admit BDD throughout the uniqueness radius $\lambda_1/2$.

Theorem 1.1 (Main result). *There are an absolute constant $C \geq 1$ and deterministic algorithms **Gen** and **ListDec** with the following properties. For every $n \geq 1$, **Gen**(1^n) outputs in time $\text{poly}(n)$ an integer basis of a full rank lattice $\hat{\mathcal{L}}_n \subseteq \mathbb{R}^n$ satisfying*

$$\sqrt{n} \leq \lambda_1(\hat{\mathcal{L}}_n) < 2\sqrt{n}, \quad \det(\hat{\mathcal{L}}_n)^{1/n} \leq C.$$

*For every $\mathbf{y} \in \mathbb{Q}^n$, **ListDec**($1^n, \mathbf{y}$) returns exactly $\{\mathbf{x} \in \hat{\mathcal{L}}_n : \|\mathbf{x} - \mathbf{y}\|_2 < \frac{\lambda_1(\hat{\mathcal{L}}_n)}{\sqrt{2}}\}$ in time $\text{poly}(n, \text{bitlen}(\mathbf{y}))$. The list has at most two elements.*

The constant C is independent of n ; we do not optimize its value. The decoder uses the public structure of the family, and all auxiliary data can be reconstructed from n in polynomial time. We compare the geometric and decoding guarantees with prior work in [Table 1](#).

1.2 Technical overview

We use Construction D [1] to combine binary codes at several distance scales. The construction requires codes with small codimension, a nested tower whose total codimension remains small, and compatible Euclidean decoders. We describe how these requirements fit together.

Codes with prescribed parameters. For every $n \geq 1$ and $1 \leq d \leq n$, we construct a binary linear code $D(n, d) \subseteq \mathbb{F}_2^n$ with

$$d(D(n, d)) \geq d, \quad \text{codim}(D(n, d)) = O\left(\sqrt{nd} \log(2n/d)\right),$$

together with a deterministic polynomial time decoder correcting every error of Hamming weight less than $d/2$.

Work	Quality $\delta/\sqrt{n}$	Dimension	Decoder	Framework / ingredients
MN08 [20] GP12 [12]	$\Theta(n^{-1/4})$	2^m	BDD: $\lambda_1/2$ List: $(1-\tau)\lambda_1$	Recursion; RM
DP19 [6]	$\Omega\left(\frac{1}{\sqrt{\log n}}\right)$	Every n	BDD: $\Theta(\sqrt{n}/\log n)$	Discrete logs
BS83 [1] MP22 [21]	$\Omega\left(\frac{1}{\sqrt{\log n}}\right)$	$2^m - 1$	List: $(1-\tau)\lambda_1/\sqrt{2}$	D; BCH
BP23 [2]	$\Omega\left(\frac{1}{\sqrt{\log n}}\right)$	Prime n	List: $\Theta(\sqrt{n}/\log n)$	A; RS
KM24 [16]	$\Omega\left(\frac{1}{(\log n)^\varepsilon \sqrt{\log \log n}}\right)$	$r^e(r-1)$	List: $(1-\tau)\lambda_1/\sqrt{2}$	D; AG
This work	$\Omega(1)$	Every n	List: $\lambda_1/\sqrt{2}$	D; BCH, expanders

Table 1. Lattice constructions with polynomial time decoding. The quality column gives a lower bound on $\delta(\mathcal{L})/\sqrt{n}$. For DP19 [6] and BP23 [2], the decoder column gives the normalized radius $R/\det(\mathcal{L})^{1/n}$; all other rows give the radius R . All listed families admit deterministic polynomial time basis construction in the stated dimensions. Radii are stated for open balls, with fixed $\tau \in (0,1)$ and $\varepsilon > 0$. The algebraic geometry construction uses its prescribed sequence $n = r^e(r-1)$ with $r = \Theta(\log n)$. A and D denote Constructions A and D; RM, RS, and AG denote Reed–Muller, Reed–Solomon, and algebraic geometry codes.

Shortened BCH codes suffice when d/n is small, and repetition codes handle constant relative distance. In the intermediate range, we use Tanner codes on spectral expanders [15,25], with shortened BCH codes as component codes. A component relative distance η gives global distance $\Omega(\eta^2 n)$ and codimension $O(n\eta \log(2/\eta))$; taking η of order $\sqrt{d/n}$ yields the required bound. The spectral expansion also ensures that alternating component decoding corrects every error of weight less than $d/2$. Graph powers provide the required expansion, and shortening gives exactly length n while preserving distance and decoding. The underlying code constructions and decoding techniques are more or less standard; we adapt them to our parameter requirements and defer the details to Section 3.

Nesting and the determinant. The codes $D_j := D(n, d_j)$ need not be nested. For a tower of depth $a \geq 1$, set $C_j := \bigcap_{t=j}^{a-1} D_t$ and $r_j := \text{codim}(D_j)$ for $0 \leq j < a$. Then $C_0 \subseteq \dots \subseteq C_{a-1}$, and $\text{codim}(C_j) \leq \sum_{t=j}^{a-1} r_t$. Choose a binary basis adapted to this chain and lift it to a unimodular integer basis $\mathbf{b}_1, \dots, \mathbf{b}_n$, ordered so that the first $k_j := \dim(C_j)$ vectors modulo two span C_j .

Construction D combines the nested codes at successive powers of two. At level j , a codeword of C_j is lifted to an integer vector using the adapted basis and scaled by 2^j . The lattice $\mathcal{L}$ consists of all such sums, together with an arbitrary vector from $2^a\mathbb{Z}^n$, thus $\mathcal{L} = \{\mathbf{x} \in \mathbb{Z}^n \mid \mathbf{x} = 2^a\mathbf{z} + \sum_{j=0}^{a-1} 2^j \sum_{i=1}^{k_j} c_{j,i} \mathbf{b}_i, \mathbf{z} \in \mathbb{Z}^n, c_{j,i} \in \{0,1\}\}$. Here the coefficients $c_{j,i}$ specify the codeword chosen from C_j in the adapted binary basis. It satisfies $\log_2 \det(\mathcal{L}) = \sum_{j=0}^{a-1} \text{codim}(C_j)$. Set $d_a := 1$ to account for the terminal lattice $2^a\mathbb{Z}^n$. The construction satisfies

$$\lambda_1(\mathcal{L})^2 \geq \min_{0 \leq j \leq a} 4^j d_j, \quad \log_2 \det(\mathcal{L}) \leq \sum_{j=0}^{a-1} (j+1) r_j. \quad (1)$$

The factor $j+1$ arises because the constraints of D_j occur in $C_0, \dots, C_j$. Taking $d_j := \lceil n/4^j \rceil$ for $0 \leq j < a$ gives $r_j = O(n(j+1)2^{-j})$. Thus the determinant bound in (1) is $O(n)$, while $4^j d_j \geq n$ at every code layer. The cost of enforcing nesting is therefore small enough to retain constant root determinant.

Euclidean decoding of the code layers. A decoder for D_j also decodes its subcode C_j : decode in D_j and check whether the result belongs to C_j . Generalized minimum-distance decoding [10,11] converts the Hamming decoder into a decoder for Euclidean distance modulo two, with radius $\sqrt{d_j}/2$. At level j , the lattice decoder recovers a parity vector in C_j , subtracts an integer lift, and divides the residual target by two. After j levels, an initial error $\mathbf{e}$ has become $\mathbf{e}/2^j$. Consequently, all code layers can be decoded uniquely whenever $\|\mathbf{e}\|_2 < \min_{0 \leq j < a} 2^j \sqrt{d_j}/2$.

List decoding at the terminal level. We choose $a := \max\{0, \lfloor \log_4(n/2) \rfloor\}$. For $a > 0$, the inclusion $2^a\mathbb{Z}^n \subseteq \mathcal{L}_n$ and (1) give $\lambda_1(\mathcal{L}_n) = 2^a$, while $2 \cdot 4^a \leq n$ ensures unique decoding at every code layer for errors of norm less than $\lambda_1(\mathcal{L}_n)/\sqrt{2}$. The remaining list is exactly $\mathbb{Z}^n \cap B_2^n(\mathbf{y}_a, 1/\sqrt{2})$, where $\mathbf{y}_a$ is the terminal target. This open ball contains at most two integer points, found by coordinatewise rounding and testing single-coordinate changes. For $a = 0$, the same enumeration applies directly to $\mathcal{L}_n = \mathbb{Z}^n$. Finally, scaling $\mathcal{L}_n$ by an integer of at most three gives the normalization in Theorem 1.1.

2 Preliminaries

Notation. Vectors and matrices are written in boldface. All vectors are columns. We write $[n] := \{1, \dots, n\}$, with $[0] = \emptyset$, and identify $\mathbb{F}_2$ with $\{0,1\}$ when taking integer representatives. Reduction modulo two is coordinatewise. The vectors $\mathbf{0}$, $\mathbf{1}$, and $\mathbf{e}_i$ denote the zero vector, the all-one vector, and the i -th standard basis vector, respectively; $\mathbf{I}_n$ is the identity matrix. We write $\|\mathbf{x}\|_2$ for the Euclidean norm and $\|\mathbf{A}\|_2 := \sup_{\|\mathbf{x}\|_2=1} \|\mathbf{A}\mathbf{x}\|_2$ for the induced matrix norm. The open Euclidean ball is $B_2^n(\mathbf{y}, R) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{y}\|_2 < R\}$. For a binary vector $\mathbf{x}$, its Hamming weight $\text{wt}(\mathbf{x})$ is the number of nonzero coordinates, and $d_H(\mathbf{x}, \mathbf{y}) := \text{wt}(\mathbf{x} - \mathbf{y})$.

For a field $\mathbb{K}$, $\text{GL}_n(\mathbb{K})$ denotes the invertible $n \times n$ matrices over $\mathbb{K}$. An integer matrix is *unimodular* if its determinant is ± 1 . These matrices form $\text{GL}_n(\mathbb{Z})$; their inverses are integral by the adjugate formula, so $\mathbf{U}\mathbb{Z}^n = \mathbb{Z}^n$ for every $\mathbf{U} \in \text{GL}_n(\mathbb{Z})$.

All running times count bit operations. For a rational vector $\mathbf{y}$, $\text{bitlen}(\mathbf{y})$ is the total binary encoding length of its numerators and denominators. All distance tests use exact arithmetic.

2.1 Binary codes and bounded distance decoding

A binary linear code of length n is a subspace $D \subseteq \mathbb{F}_2^n$; we use standard coding terminology as in [17]. Its minimum distance and codimension are

$$d(D) := \min_{\mathbf{0} \neq \mathbf{c} \in D} \text{wt}(\mathbf{c}), \quad \text{codim}(D) := n - \dim(D),$$

with $d(\{\mathbf{0}\}) = +\infty$. A generator matrix has a basis of D as its columns. A parity check matrix $\mathbf{H}$ satisfies $D = \ker \mathbf{H}$; row reduction gives a full row rank parity check matrix with $\text{codim}(D)$ rows.

Definition 2.1 (Bounded distance decoding). *Let $d \geq 1$ be an integer and suppose $d(D) \geq d$. Set $T_d := \lfloor (d-1)/2 \rfloor$. A bounded distance decoder with radius T_d takes $\mathbf{z} \in \mathbb{F}_2^n$ and returns the unique codeword $\mathbf{c} \in D$ satisfying $d_H(\mathbf{z}, \mathbf{c}) \leq T_d$, if one exists, and returns $\perp$ otherwise.*

Uniqueness follows from $2T_d < d$. A decoder guaranteed to recover such a codeword can be made to satisfy the failure condition by checking membership and distance. We use this convention for all bounded distance decoders.

Lemma 2.2. *For binary linear codes $D_1, \dots, D_m$ of the same length,*

$$\text{codim}\left(\bigcap_{i=1}^m D_i\right) \leq \sum_{i=1}^m \text{codim}(D_i).$$

The intersection can be constructed by stacking parity check matrices.

Proof. The intersection is the kernel of the stacked matrix, whose rank is at most the sum of the individual ranks.

Shortening on a set of coordinates means retaining the codewords that vanish on those coordinates and then deleting those coordinates [17].

Lemma 2.3 (Shortening). *Shortening a binary linear code $D' \subseteq \mathbb{F}_2^{n'}$ on any $n' - n$ coordinates gives a code $D \subseteq \mathbb{F}_2^n$ satisfying $d(D) \geq d(D')$ and $\text{codim}(D) \leq \text{codim}(D')$. Its matrices can be computed in polynomial time. A bounded distance decoder with radius T_d for D' induces one for D with polynomially related running time.*

Proof. Deleting zero coordinates preserves Hamming weight. The zero constraints reduce dimension by at most $n' - n$, so $\text{codim}(D) \leq \text{codim}(D')$. A parity-check matrix for D is obtained by deleting the corresponding columns; a generator matrix follows by linear algebra.

To decode, extend the input by zeros and decode in D' . If the result vanishes on the added coordinates and is within distance T_d , delete those coordinates and return it; otherwise return $\perp$. Correctness follows from the distance preserving correspondence between D and the codewords of D' vanishing on those coordinates.

2.2 Lattices and list decoding

A full-rank lattice in $\mathbb{R}^n$ is a set $\mathcal{L} = \mathbf{B}\mathbb{Z}^n$ for an invertible real matrix $\mathbf{B}$, called a basis of $\mathcal{L}$. Its determinant and minimum distance are $\det(\mathcal{L}) := |\det \mathbf{B}|$, $\lambda_1(\mathcal{L}) := \min_{\mathbf{0} \neq \mathbf{x} \in \mathcal{L}} \|\mathbf{x}\|_2$. For a target $\mathbf{y} \in \mathbb{R}^n$, write the distance from $\mathbf{y}$ to $\mathcal{L}$ as $\text{dist}(\mathbf{y}, \mathcal{L}) := \min_{\mathbf{x} \in \mathcal{L}} \|\mathbf{y} - \mathbf{x}\|_2$. The normalized minimum distance and Hermite invariant are defined as $\delta(\mathcal{L}) := \lambda_1(\mathcal{L}) / \det(\mathcal{L})^{1/n}$ and $\gamma(\mathcal{L}) := \delta(\mathcal{L})^2$. Scaling a lattice by $s > 0$ multiplies both $\lambda_1(\mathcal{L})$ and $\det(\mathcal{L})^{1/n}$ by s , so $\delta(\mathcal{L})$ and $\gamma(\mathcal{L})$ are unchanged. All lattices constructed here have integer bases.

For $R > 0$, define $\text{List}(\mathcal{L}, \mathbf{y}, R) := \{\mathbf{x} \in \mathcal{L} : \|\mathbf{x} - \mathbf{y}\|_2 < R\}$. A list decoder returns exactly this set; its running time includes writing the output. If $\text{dist}(\mathbf{y}, \mathcal{L}) < \lambda_1(\mathcal{L})/2$, the closest lattice vector is unique by the triangle inequality. Thus a list decoder with $R \geq \lambda_1(\mathcal{L})/2$ also gives bounded distance decoding under this promise, by choosing a closest point in the list. The strict inequality is necessary: if $\mathbf{v}$ is a shortest nonzero lattice vector, then $\mathbf{v}/2$ is equally close to $\mathbf{0}$ and $\mathbf{v}$. For rational targets and integer lattices, squared distances are rational and can be compared exactly.

2.3 Divisibility and parity

For a nonzero integer z , its *2-adic valuation* is

$$\nu_2(z) := \max\{q \in \mathbb{Z}_{\geq 0} : 2^q \mid z\}, \quad \nu_2(0) := +\infty.$$

Thus $z = 2^{\nu_2(z)}u$ for an odd integer u . For $\mathbf{0} \neq \mathbf{x} \in \mathbb{Z}^n$, set $q := \min_i \nu_2(x_i)$. Then $\mathbf{u} := 2^{-q}\mathbf{x} \in \mathbb{Z}^n$ and $\mathbf{u} \bmod 2 \neq \mathbf{0}$. Since every odd coordinate has absolute value at least one, $\|\mathbf{u}\|_2^2 \geq \text{wt}(\mathbf{u} \bmod 2)$. Consequently, if $\mathbf{u} \bmod 2$ belongs to a binary code of distance at least d , then $\|\mathbf{x}\|_2^2 = 4^q \|\mathbf{u}\|_2^2 \geq 4^q d$. This relates the code distance to the Euclidean norm.

Unimodular matrices preserve divisibility: for $\mathbf{U} \in \text{GL}_n(\mathbb{Z})$ and $q \geq 0$, $\mathbf{U}(2^q\mathbb{Z}^n) = 2^q\mathbb{Z}^n$, since $\mathbf{U}\mathbb{Z}^n = \mathbb{Z}^n$. In particular, $\mathbf{x} \in 2^q\mathbb{Z}^n$ if and only if $\mathbf{U}\mathbf{x} \in 2^q\mathbb{Z}^n$.

2.4 Explicit expanders and the expander mixing lemma

We follow the terminology of Vadhan [29, Chapter 4]. Graphs may have parallel edges and self-loops. At each vertex of a Δ -regular graph, the incident edge ends

are labelled by $[\Delta]$; these labels are also called ports. The *edge rotation map* is an involution

$$\text{Rot}_X : V \times [\Delta] \longrightarrow V \times [\Delta].$$

The equality $\text{Rot}_X(v, i) = (w, j)$ means that the edge with label i at v has label j at its other endpoint w . We use the rotation map convention for self-loops: a fixed point contributes one port, while two distinct ports paired at the same vertex contribute two [25, Definition 2.1]. All degrees and adjacency entries count these multiplicities.

The adjacency matrix of X is the symmetric matrix

$$\mathbf{A}_X(v, w) := |\{i : \text{Rot}_X(v, i) = (w, j) \text{ for some } j\}|.$$

Every row has sum Δ . The *random walk matrix* is $\mathbf{P}_X := \mathbf{A}_X/\Delta$; its (v, w) entry is the probability of moving from v to w by choosing a port uniformly. For $M := |V|$ and $\mathbf{J}_M := \mathbf{1}\mathbf{1}^\top/M$, define

$$\lambda(X) := \|\mathbf{P}_X - \mathbf{J}_M\|_2 = \|\mathbf{P}_X|_{\mathbf{1}^\perp}\|_2.$$

Since $\mathbf{P}_X$ is symmetric, $\lambda(X)$ is the largest absolute eigenvalue on $\mathbf{1}^\perp$. The *spectral expansion* is $1 - \lambda(X)$ in the terminology of [29, Definition 4.5].

A constant degree graph family is *fully explicit* if the neighbor with a specified port can be computed in time polynomial in the bit length of the vertex label [29, Section 4.3]. The following construction also computes the return port.

Theorem 2.4 (Explicit constant degree expanders [25, Theorem 3.3]). *There are absolute integers $b \geq 2$, $\Delta_0 \geq 3$, a rational constant $0 < \rho < 1$, and a fully explicit family $\{X_t\}_{t \geq 1}$ such that X_t has b^t vertices, degree Δ_0 , and $\lambda(X_t) \leq \rho$. Its edge rotation map is computable in $\text{poly}(t)$ time. The base graph and all three constants are fixed independently of subsequent parameters.*

This follows from [25, Theorem 3.3], with a fixed base graph supplied by [25, Section 5.1]; see also [29, Theorem 4.39].

For an integer $s \geq 1$, the *graph power* X^s is defined by length- s walks. Its ports at each vertex are indexed by s -tuples $(i_1, \dots, i_s)$ of base ports. If $\text{Rot}_X(v_{r-1}, i_r) = (v_r, j_r)$ for $1 \leq r \leq s$, then

$$\text{Rot}_{X^s}(v_0, (i_1, \dots, i_s)) = (v_s, (j_s, \dots, j_1)).$$

Reversing the return labels makes this map an involution.

Lemma 2.5 (Graph powers [25, Proposition 2.3]). *If X has degree Δ and $\lambda(X) \leq \rho$, then X^s has degree Δ^s , random walk matrix $\mathbf{P}_{X^s}$, and $\lambda(X^s) \leq \rho^s$. Its edge rotation map uses s calls to that of X , with additional time polynomial in the label lengths.*

The *bipartite double cover* $\text{Bip}(X)$ has left and right copies of $V(X)$. For every port (v, i) with $\text{Rot}_X(v, i) = (w, j)$, place an edge from v_L to w_R , with label i at v_L and label j at w_R . Since the rotation map is a bijection, every

right port occurs once. Thus the double cover is Δ -regular on each side, has $M\Delta$ edges, and has biadjacency matrix $\mathbf{A}_X$. The two ports of an edge joining distinct vertices in X give two distinct edges in the double cover.

More generally, let $\Gamma = (V_L, V_R, E)$ be a Δ -regular bipartite multigraph with M vertices on each side. Its biadjacency matrix has rows indexed by V_L , columns indexed by V_R , and entries counting edges between the corresponding vertices. Let $\mathbf{Q}_\Gamma$ be this matrix divided by Δ , and define

$$\sigma(\Gamma) := \|\mathbf{Q}_\Gamma - \mathbf{J}_M\|_2.$$

Both the row sums and the column sums of $\mathbf{Q}_\Gamma$ are one, so $\sigma(\Gamma)$ is its largest non-trivial singular value. In particular, $\sigma(\text{Bip}(X)) = \lambda(X)$. This parameter refers to the biadjacency matrix; the full random walk matrix of a bipartite graph has eigenvalue -1 . For $S \subseteq V_L$ and $T \subseteq V_R$, let $e(S, T)$ count the edges with one endpoint in S and the other in T , counting parallel edges separately.

Lemma 2.6 (Expander mixing lemma [29, Lemma 4.15]). *If $\sigma(\Gamma) \leq \theta$, then for every $S \subseteq V_L$ and $T \subseteq V_R$,*

$$\left| e(S, T) - \frac{\Delta |S| |T|}{M} \right| \leq \theta \Delta \sqrt{|S| |T|}.$$

This is the directed version of the cited lemma, applied after identifying each vertex class with $[M]$ and viewing the biadjacency matrix as the adjacency matrix of a directed graph.

3 Codes and Their Decoders

We construct binary codes of every prescribed length and designed distance, together with their bounded distance decoders. We then extend these decoders to Euclidean distance modulo two, as required by the lattice decoder in [Section 4.3](#).

3.1 Shortened binary BCH codes

We first recall the classical construction of shortened binary BCH codes [14, 3, 17]. The construction and decoder are standard. We include the proof to specify them for every prescribed length and designed distance. These codes will be used both as component codes in the Tanner construction and directly when the required distance is small.

Theorem 3.1 (Shortened BCH codes [17, 23]). *For every $N \geq 1$ and $1 \leq h \leq N$, a binary linear code $B(N, h) \subseteq \mathbb{F}_2^N$ can be constructed deterministically in $\text{poly}(N)$ time, with*

$$d(B(N, h)) \geq h, \quad \text{codim}(B(N, h)) \leq (h - 1) \lceil \log_2(N + 1) \rceil.$$

The code has a deterministic bounded distance decoder with radius T_h running in $\text{poly}(N)$ time.

Proof. Set $m = \lceil \log_2(N+1) \rceil$ and $\mathbb{K} = \mathbb{F}_{2^m}$. Choose distinct nonzero elements $x_1, \dots, x_N \in \mathbb{K}$ and define $B(N, h) := \left\{ \mathbf{c} \in \mathbb{F}_2^N : \sum_{j=1}^N c_j x_j^r = 0, \forall r \in [h-1] \right\}$. With coordinates indexed by all of $\mathbb{K}^\times$, these are the parity checks of a primitive narrow-sense binary BCH code of designed distance h , up to coordinate ordering. The code $B(N, h)$ is obtained by shortening on the unused coordinates. When $h = 1$, there are no constraints and $B(N, 1) = \mathbb{F}_2^N$.

Suppose a nonzero codeword $\mathbf{c}$ has weight $w < h$, with support $J = \{j_1, \dots, j_w\}$. The first w parity checks give $\mathbf{V}\mathbf{c}_J = \mathbf{0}$, where $\mathbf{V} = (x_{j_s}^r)_{1 \leq r, s \leq w}$, $\det(\mathbf{V}) = (\prod_{s=1}^w x_{j_s}) \prod_{1 \leq a < b \leq w} (x_{j_b} - x_{j_a}) \neq 0$. This contradicts $\mathbf{c}_J \neq \mathbf{0}$, proving the distance bound. Each equation over $\mathbb{K}$ gives at most m binary equations, so $\text{codim}(B(N, h)) \leq (h-1)m$.

The classical BCH syndrome decoder corrects every error pattern of weight at most $T_h = \lfloor (h-1)/2 \rfloor$ in time polynomial in 2^m [23, 19] is given in Appendix A. Extend the received word by zeros in the shortened coordinates and apply this decoder. Accept only a codeword that vanishes in those coordinates and lies within distance T_h of the extended input. By Lemma 2.3, this gives the required decoder for $B(N, h)$. For $h = 1$, simply return the input.

Finally, enumerate the monic binary polynomials of degree m in lexicographic order until an irreducible polynomial f is found. Each candidate can be tested deterministically for irreducibility in $\text{poly}(m)$ time [24]. Since there are $2^m = O(N)$ candidates, this takes $\text{poly}(N)$ time. Represent $\mathbb{K}$ as $\mathbb{F}_2[X]/(f)$ and order its nonzero elements lexicographically by their coefficient vectors. Field operations take $\text{poly}(m)$ bit operations. Thus the field, the binary parity check matrix, and the decoding data can all be constructed deterministically in $\text{poly}(N)$ time, and decoding has polynomial bit complexity.

3.2 Tanner codes on bipartite graphs

Tanner introduced codes defined by local constraints on a graph [28]. We use the bipartite construction studied by Zémor [33] and Janwa and Lal [15], in which each edge carries one binary coordinate and each vertex imposes a code constraint on its incident edges.

Let $M, \Delta \geq 1$ be integers, and let $\Gamma = (V_L, V_R, E)$ be a Δ -regular bipartite multigraph with disjoint vertex sets V_L, V_R of size M . Here E is the edge set: every edge has one endpoint in V_L and one in V_R , and parallel edges are distinct elements of E . Thus $|E| = M\Delta$. A word assigns one bit $x_e \in \mathbb{F}_2$ to each edge $e \in E$. We denote the set of all such assignments by $\mathbb{F}_2^E$. After ordering the edges as $e_1, \dots, e_{|E|}$, each assignment is represented by the vector $(x_{e_1}, \dots, x_{e_{|E|}})^\top \in \mathbb{F}_2^{|E|}$.

For each vertex v , let $E(v) \subseteq E$ be its set of incident edges. Since $|E(v)| = \Delta$, we can fix an ordering represented by a bijection $\iota_v : [\Delta] \rightarrow E(v)$. Define $\pi_v : \mathbb{F}_2^E \rightarrow \mathbb{F}_2^\Delta$ by $\pi_v(\mathbf{x}) := (x_{\iota_v(1)}, \dots, x_{\iota_v(\Delta)})^\top$. Thus $\pi_v(\mathbf{x})$ consists of the coordinates on the edges incident to v , in the specified order.

For a binary linear code $B \subseteq \mathbb{F}_2^\Delta$, define the *Tanner code* $\mathcal{T}(\Gamma, B) := \{\mathbf{x} \in \mathbb{F}_2^E : \pi_v(\mathbf{x}) \in B \text{ for every } v \in V_L \cup V_R\}$. The code B is called the *component code*,

because it constrains the coordinates incident to each vertex. The local orderings are part of the construction and remain fixed throughout decoding. If $\mathbf{H}_B$ is a full row rank parity check matrix of B , then $\mathcal{T}(\Gamma, B) = \{\mathbf{x} \in \mathbb{F}_2^E : \mathbf{H}_B \cdot \pi_v(\mathbf{x}) = \mathbf{0} \text{ for every } v \in V_L \cup V_R\}$. These are homogeneous binary linear equations, so $\mathcal{T}(\Gamma, B)$ is a binary linear code. Throughout this subsection, assume $d(B) \geq h$ for an integer $1 \leq h \leq \Delta$. We use $\sigma(\Gamma)$ as defined in [Section 2.4](#); for $S \subseteq V_L$ and $T \subseteq V_R$, $e(S, T)$ counts the edges joining S to T , including multiplicity.

For equal component codes, the following distance bound follows from [\[15, Theorem 3.1\]](#). We include the proof with our normalization; it also covers parallel edges.

Lemma 3.2. *Suppose $\sigma(\Gamma) \leq \theta$, $0 < \eta \leq 1$, and $h \geq \eta\Delta$ with $\theta \leq \eta/4$. Then $\text{codim}(\mathcal{T}(\Gamma, B)) \leq 2M \text{codim}(B)$, and $d(\mathcal{T}(\Gamma, B)) \geq \frac{3}{4}\eta^2 |E|$.*

Proof. At each vertex, the constraint $\mathbf{H}_B \pi_v(\mathbf{x}) = \mathbf{0}$ imposes at most $\text{rank}(\mathbf{H}_B) = \text{codim}(B)$ independent linear equations. Stacking these equations over the $2M$ vertices gives the codimension bound.

For the distance bound, let $F \subseteq E$ be the support of a nonzero codeword, and define $S := \{v \in V_L : E(v) \cap F \neq \emptyset\}$, $T := \{v \in V_R : E(v) \cap F \neq \emptyset\}$. At every vertex in $S \cup T$, the local restriction is a nonzero codeword of B and therefore has weight at least $h \geq \eta\Delta$. Counting the edges in F on each side gives $|S|, |T| \leq \frac{|F|}{\eta\Delta}$. Every edge in F joins S to T . Hence [Lemma 2.6](#) implies

$$\begin{aligned} |F| &\leq e(S, T) \leq \frac{\Delta |S| |T|}{M} + \theta \Delta \sqrt{|S| |T|} \\ &\leq \frac{|F|^2}{\eta^2 |E|} + \frac{\theta}{\eta} |F| \leq \frac{|F|^2}{\eta^2 |E|} + \frac{1}{4} |F|. \end{aligned}$$

Since $|F| > 0$, rearranging gives $|F| \geq 3\eta^2 |E|/4$.

We next give a variant of Zémor's alternating decoder [\[33, Section III-A\]](#) using a bounded distance decoder for the component code B . Let Dec_B be deterministic and have radius $T_h := \lfloor \frac{h-1}{2} \rfloor$. Given $\mathbf{z} \in \mathbb{F}_2^\Delta$, it returns the unique codeword within distance T_h , if one exists, and returns $\perp$ otherwise. Define the

update $f : \mathbb{F}_2^\Delta \rightarrow \mathbb{F}_2^\Delta$ by $f(\mathbf{z}) := \begin{cases} \text{Dec}_B(\mathbf{z}), & \text{Dec}_B(\mathbf{z}) \neq \perp, \\ \mathbf{z}, & \text{Dec}_B(\mathbf{z}) = \perp. \end{cases}$ At a vertex v ,

we apply f to $\pi_v(\mathbf{x})$ and write the resulting bits back to the incident edges, in the fixed order given by ι_v . Thus a successful component decoding replaces the incident coordinates by a codeword of B , while a failed decoding leaves them unchanged.

For either side $W \in \{V_L, V_R\}$, every edge has exactly one endpoint in W , so the sets $\{E(v) : v \in W\}$ partition E . We can therefore update all vertices in W simultaneously, with each edge updated exactly once. Define $F_W : \mathbb{F}_2^E \rightarrow \mathbb{F}_2^E$ by $\pi_v(F_W(\mathbf{x})) = f(\pi_v(\mathbf{x}))$, $v \in W$. These identities uniquely determine $F_W(\mathbf{x})$. Starting from the received word $\mathbf{y}$, the algorithm applies $F_{V_L}, F_{V_R}, F_{V_L}, \dots$ to the current word in alternating order. One update of an entire side is called a *half-round*.

Lemma 3.3. Define $\beta := \frac{T_h+1}{\Delta} = \frac{\lceil h/2 \rceil}{\Delta}$ and assume $\sigma(\Gamma) \leq \theta < \beta$. For every $\mathbf{c}^* \in \mathcal{T}(\Gamma, B)$ and $\mathbf{y} \in \mathbb{F}_2^E$ satisfying $d_H(\mathbf{y}, \mathbf{c}^*) < \beta(\beta - \theta)|E|$, the alternating algorithm produces $\mathbf{c}^*$ after $M + 1$ half-rounds.

For an integer $1 \leq d_0 \leq d(\mathcal{T}(\Gamma, B))$ with $T_{d_0} < \beta(\beta - \theta)|E|$, accept the resulting word $\mathbf{x}$ only if $\mathbf{x} \in \mathcal{T}(\Gamma, B)$ and $d_H(\mathbf{x}, \mathbf{y}) \leq T_{d_0}$; otherwise return $\perp$. This gives a deterministic bounded distance decoder with radius T_{d_0} . Given the graph's incidence lists, the fixed coordinate orderings, and a full row rank parity check matrix of B , its running time is $\text{poly}(|E|)$ whenever Dec_B runs in $\text{poly}(\Delta)$ time.

Proof. Fix $\mathbf{c}^* \in \mathcal{T}(\Gamma, B)$. Throughout the proof, an edge e is erroneous if its current bit differs from c_e^* . For every vertex v and $\mathbf{z} \in \mathbb{F}_2^\Delta$, the bounded distance guarantee gives $d_H(\mathbf{z}, \pi_v(\mathbf{c}^*)) \leq T_h \implies f(\mathbf{z}) = \pi_v(\mathbf{c}^*)$. Consequently, if the restriction at v is incorrect after an update, at least $T_h + 1 = \beta\Delta$ edges incident to v were erroneous before that update.

After the first left half-round, let $\mathbf{x}'$ be the current word and define $S := \{v \in V_L : \pi_v(\mathbf{x}') \neq \pi_v(\mathbf{c}^*)\}$. Each vertex in S was incident to at least $\beta\Delta$ edges e with $y_e \neq c_e^*$. Since each edge has exactly one left endpoint, $\beta\Delta|S| \leq d_H(\mathbf{y}, \mathbf{c}^*)$. By the definition of S , every edge e with $x'_e \neq c_e^*$ has its left endpoint in S .

After the next right half-round, let $\mathbf{x}''$ be the current word and define $T := \{v \in V_R : \pi_v(\mathbf{x}'') \neq \pi_v(\mathbf{c}^*)\}$. Before this update, each vertex in T was incident to at least $\beta\Delta$ edges e with $x'_e \neq c_e^*$. The left endpoint of every such edge lies in S , so $\beta\Delta|T| \leq e(S, T)$. Applying Lemma 2.6 and dividing by Δ gives $\beta|T| \leq \frac{|S||T|}{M} + \theta\sqrt{|S||T|}$. If $|T| \geq |S| > 0$, then $\sqrt{|S||T|} \leq |T|$. Dividing by $|T|$ therefore yields $\beta \leq |S|/M + \theta$. Hence $0 < |S| < (\beta - \theta)M$ implies $|T| < |S|$. By the assumed error bound and $|E| = M\Delta$, $|S| \leq \frac{d_H(\mathbf{y}, \mathbf{c}^*)}{\beta\Delta} < (\beta - \theta)M$. If S is empty, every edge is already correct. Otherwise, $|T| < |S|$, so the same argument applies to the next half-round with the two sides exchanged. Repeating this argument, the number of vertices with incorrect restrictions on the most recently processed side strictly decreases until it is zero. This number is initially at most M , so at most M further half-rounds are needed. Once the current word equals $\mathbf{c}^*$, every subsequent update leaves it unchanged.

For the bounded distance decoder, any codeword within distance T_{d_0} of the original input is recovered, since $T_{d_0} < \beta(\beta - \theta)|E|$, and therefore passes both final tests. Conversely, every accepted output belongs to $\mathcal{T}(\Gamma, B)$ and lies within distance T_{d_0} of the original input and there is at most one such codeword because $2T_{d_0} < d(\mathcal{T}(\Gamma, B))$.

There are $M + 1$ half-rounds, each using M calls to the component decoder. The final membership test checks $\mathbf{H}_B \pi_v(\mathbf{x}) = \mathbf{0}$ at every vertex, and the distance test examines the $|E|$ coordinates. Since $M, \Delta \leq |E|$, the total running time is $\text{poly}(|E|)$.

The above proof follows the contraction argument in [33, Lemma 5 and Theorem 6]; see also [15, Section 5].

3.3 Codes with prescribed length and distance

We now combine the BCH and Tanner constructions. For every prescribed length n and designed distance d , the resulting code has the codimension bound needed by the lattice construction and a bounded distance decoder with radius T_d .

Parameter choices. For integers $n \geq 1$ and $1 \leq d \leq n$, define $\delta := d/n$ and $m(n) := \lceil \log_2(n+2) \rceil$. We write $m = m(n)$ when n is clear. The construction uses a repetition code when $\delta \geq 2^{-8}$, a shortened BCH code when $\delta < 2^{-8}$ and $\delta \leq m^{-4}$, and a Tanner code in the remaining range $m^{-4} < \delta < 2^{-8}$.

For the Tanner construction, let b, Δ_0, ρ be the constants in [Theorem 2.4](#), and choose the following parameters in the indicated order. The parameters t, M are needed only when $\Delta \leq n$.

Parameter	Description	Choice
η	Required relative distance of the component code $B := B(\Delta, h)$.	$\eta := 2\sqrt{\delta}$, thus $2m^{-2} < \eta < 1/8$.
s, Δ	Walk length in X_t ; degree of $\Gamma := \text{Bip}(X_t^s)$ and length of B .	Choose the least integer $s \geq 1$ s.t. $\Delta_0^s \geq \eta^{-2}$, $\rho^s \leq \eta/4$, and set $\Delta := \Delta_0^s$.
t, M	Index of the expander X_t ; number $ V_L = V_R $ of vertices on each side of Γ .	When $\Delta \leq n$, choose the least integer $t \geq 1$ s.t. $b^t \Delta \geq n$, and set $M := b^t$. Then $n \leq M\Delta \leq bn$.
h	Designed distance of the component code B .	$h := \lceil \eta\Delta \rceil$, so $\eta\Delta \leq h \leq \Delta$.
θ	Upper bound on $\sigma(\Gamma)$.	$\theta := \rho^s \leq \eta/4$.

When $\Delta \leq n$, the Tanner construction uses $\Gamma := \text{Bip}(X_t^s) = (V_L, V_R, E)$, with M vertices on each side, degree Δ , and $|E| = M\Delta$ edges. The choice of s gives $s = O(\log(2/\eta))$ and $\Delta \leq (2/\eta)^{O(1)} \leq m^{O(1)}$.

All parameter choices use integer arithmetic. The condition $\delta \leq m^{-4}$ is equivalent to $dm^4 \leq n$. Writing $\rho = p/q$, choose s by testing $4d\Delta_0^s \geq n$ and $4np^{2s} \leq dq^{2s}$. Similarly, h is the least positive integer satisfying $nh^2 \geq 4d\Delta^2$.

Theorem 3.4 (Codes with prescribed length and distance). *There is an absolute constant $A > 0$ and a deterministic algorithm that, given integers $n \geq 1$ and $1 \leq d \leq n$, constructs a binary linear code $D(n, d) \subseteq \mathbb{F}_2^n$ satisfying*

$$d(D(n, d)) \geq d, \quad \text{codim}(D(n, d)) \leq A\sqrt{nd} \log_2 \left(\frac{2n}{d} \right). \quad (2)$$

The algorithm outputs a parity check matrix and decoding data in $\text{poly}(n)$ time. The code admits a deterministic bounded distance decoder with radius T_d running in $\text{poly}(n)$ time.

Proof. Define $\delta := d/n$ and write $m := m(n) = \lceil \log_2(n+2) \rceil$. If $\delta \geq 2^{-8}$, take $D(n, d) = \{\mathbf{0}, \mathbf{1}\} \subseteq \mathbb{F}_2^n$. This code has distance n and codimension $n-1$. Since $\sqrt{\delta} \log_2(2/\delta) \geq 1/16$, (2) holds for $A \geq 16$. Decode by majority, breaking ties deterministically, and accept the chosen codeword only if its distance from the input is at most T_d .

If $\delta < 2^{-8}$ and $\delta \leq m^{-4}$, take $D(n, d) = B(n, d)$ from Theorem 3.1. Since $\sqrt{\delta} m \leq 1$, we have $\text{codim}(D(n, d)) \leq dm = n\sqrt{\delta}(\sqrt{\delta} m) \leq n\sqrt{\delta}$. This implies (2), because $\log_2(2/\delta) \geq 1$. The distance and decoding guarantees follow from Theorem 3.1.

It remains to consider $m^{-4} < \delta < 2^{-8}$. Define $\eta := 2\sqrt{\delta}$, so $2m^{-2} < \eta < 1/8$. Let b, Δ_0, ρ be the constants in Theorem 2.4. Choose the least integer $s \geq 1$ such that $\Delta_0^s \geq \eta^{-2}$ and $\rho^s \leq \eta/4$, and set $\Delta := \Delta_0^s$. Then $s = O(\log(2/\eta))$, so $\Delta \leq (2/\eta)^{O(1)} \leq m^{O(1)}$. If $\Delta > n$, take $D(n, d) = \{\mathbf{0}\}$. Its decoder returns $\mathbf{0}$ if the input has weight at most T_d , and $\perp$ otherwise. Since $\Delta \leq m^{O(1)}$ and $m = O(\log(n+2))$, this case occurs for only finitely many pairs (n, d) , so increasing A covers their codimension n .

Assume henceforth that $\Delta \leq n$. Choose the least integer $t \geq 1$ such that $b^t \Delta \geq n$, set $M := b^t$, and let $\Gamma := \text{Bip}(X_t^s) = (V_L, V_R, E)$. This graph has M vertices on each side, degree Δ , and $|E| = M\Delta$ edges. If $t = 1$, then $|E| = b\Delta \leq bn$; otherwise, minimality gives $b^{t-1}\Delta < n$. Thus $n \leq |E| \leq bn$.

At each vertex, order the length- s tuples of base edge labels lexicographically. The rotation map specifies the corresponding incident edges and their return labels, thereby fixing the bijections $\iota_v : [\Delta] \rightarrow E(v)$. Order the coordinates by the left endpoint of each edge and then by its edge label at that endpoint. Define $\theta := \rho^s$. By Lemma 2.5 and the definition of Γ , $\sigma(\Gamma) \leq \theta \leq \eta/4$.

Set $h := \lceil \eta\Delta \rceil$ and use the component code $B := B(\Delta, h)$. Since $\eta < 1/8$ and $\Delta \geq \eta^{-2} > 64$, we have $1 \leq h \leq \Delta$. By Theorem 3.1, $d(B) \geq h \geq \eta\Delta$ and $\text{codim}(B) \leq (h-1) \lceil \log_2(\Delta+1) \rceil < \eta\Delta \log_2(2\Delta) = O(\eta\Delta \log_2(2/\eta))$. Here we used $h-1 < \eta\Delta$, $\lceil \log_2(\Delta+1) \rceil \leq \log_2(2\Delta)$, and $s = O(\log(2/\eta))$.

Let $D' := \mathcal{T}(\Gamma, B) \subseteq \mathbb{F}_2^{|E|}$ under the specified coordinate ordering. By Lemma 3.2, and $n \leq |E| \leq bn$, we have $\text{codim}(D') \leq 2M \text{codim}(B) = O(n\sqrt{\delta} \log_2(2/\delta))$ and $d(D') \geq \frac{3}{4}\eta^2 |E| = 3\delta |E| \geq 3d$.

For decoding, define $\beta := \lceil h/2 \rceil / \Delta$. Then $\beta \geq \eta/2$ and $\beta - \theta \geq \eta/4$, so $\beta(\beta - \theta) |E| \geq \eta^2 |E| / 8 = \delta |E| / 2 \geq d/2 > T_d$. The component decoder is supplied by Theorem 3.1. Thus Lemma 3.3, with $d_0 = d$, gives a deterministic bounded distance decoder for D' with radius T_d .

Shorten D' on its final $|E| - n$ coordinates, and denote the resulting code by $D(n, d)$. By Lemma 2.3, $d(D(n, d)) \geq d(D')$, $\text{codim}(D(n, d)) \leq \text{codim}(D')$, and $D(n, d)$ admits a deterministic polynomial-time bounded distance decoder with radius T_d . The bounds on $\text{codim}(D')$ and $d(D')$ imply (2) for a sufficiently large absolute A .

In the Tanner case, $s, t = O(\log(n+2))$ and $|E| \leq bn$, so the rotation maps allow us to enumerate all edges and their coordinate orderings in $\text{poly}(n)$ time. The component code and its decoder are constructible in $\text{poly}(\Delta)$ time. Stacking the component parity checks at all vertices and performing shortening

use polynomial-time binary linear algebra, and the alternating decoder runs in $\text{poly}(n)$ time by [Lemma 3.3](#). All stored data have polynomial bit length. Together with the repetition, BCH, and zero-code cases, this proves the construction and decoding claims.

Remark 3.5 (Alternative code constructions). The guarantees of [Theorem 3.4](#) can also be obtained from classical concatenated codes [9]. With suitable parameters and shortening, Reed–Solomon outer codes and binary linear inner codes of logarithmic length give the required distance and codimension bounds. The inner codes can be constructed by the linear Gilbert–Varshamov argument [17] and decoded by exhaustive search, both in deterministic polynomial time. Forney’s generalized minimum-distance decoding [10] corrects every error pattern of weight less than half the product of the component designed distances, giving the required bounded distance decoder. Our lattice construction uses only the guarantees of [Theorem 3.4](#) and therefore applies to either approach. In this paper, we give the details only for the expander-based instantiation, using shortened BCH codes as component codes.

3.4 Generalized minimum-distance decoding

The lattice decoder receives rational vectors rather than binary words. We use generalized minimum-distance (GMD) decoding [10,11] to pass from Hamming distance to Euclidean distance modulo two. For $\mathbf{y} \in \mathbb{R}^N$ and $\mathbf{c} \in \mathbb{F}_2^N$, define $d_{2,\text{tor}}(\mathbf{y}, \mathbf{c}) := \min_{\mathbf{q} \in \mathbb{Z}^N} \|\mathbf{y} - \mathbf{c} - 2\mathbf{q}\|_2$.

Lemma 3.6 (Generalized minimum-distance decoding). *Let $D \subseteq \mathbb{F}_2^N$ be a code of minimum Hamming distance at least d , where $d \geq 1$ is an integer, and suppose that D has a deterministic bounded distance decoder with radius $T_d = \lfloor (d-1)/2 \rfloor$. There is a deterministic algorithm that, given $\mathbf{y} \in \mathbb{Q}^N$, returns the unique codeword $\mathbf{c} \in D$ satisfying $d_{2,\text{tor}}(\mathbf{y}, \mathbf{c}) < \sqrt{d}/2$, if one exists, and returns $\perp$ otherwise. The algorithm uses at most $2(N+1)$ calls to the bounded distance decoder, with overhead polynomial in N and $\text{bitlen}(\mathbf{y})$.*

Proof. For each coordinate, let $p_i \in \mathbb{F}_2$ be the parity of a nearest integer to y_i , with ties broken deterministically, and put $t_i := \text{dist}(y_i, \mathbb{Z}) \in [0, 1/2]$. The distances from y_i to the parity classes $p_i + 2\mathbb{Z}$ and $(1-p_i) + 2\mathbb{Z}$ are t_i and $1-t_i$, respectively. Thus, when t_i is close to $1/2$, the two parity choices have nearly equal distances.

The algorithm works as follows. Sort the distinct values among $0, t_1, \dots, t_N, \frac{1}{2}$, and take the midpoint of each consecutive interval as a threshold τ . For each threshold, mark the coordinates in $\mathcal{E}_\tau := \{i : t_i \geq \tau\}$ as erased. For each $b \in \{0, 1\}$, form $\mathbf{z}_\tau^{(b)} \in \mathbb{F}_2^N$ by setting every erased coordinate to b and retaining p_i at every other coordinate. Apply the bounded distance decoder to both words and return any resulting codeword $\mathbf{c}$ satisfying $d_{2,\text{tor}}(\mathbf{y}, \mathbf{c})^2 < d/4$. If no candidate passes this test, return $\perp$.

To prove recovery, fix a codeword $\mathbf{c} \in D$ satisfying $d_{2,\text{tor}}(\mathbf{y}, \mathbf{c})^2 < d/4$. For a threshold τ , let e_τ count the coordinates outside $\mathcal{E}_\tau$ where the rounded bit

$p_i \neq c_i$. This error count is used only in the analysis; the algorithm does not know $\mathbf{c}$ or compute e_τ .

It suffices to find an enumerated threshold satisfying $2e_\tau + |\mathcal{E}_\tau| < d$. Indeed, at each erased coordinate, exactly one of the all-zero and all-one fillings differs from c_i . The two fillings therefore have a total of $|\mathcal{E}_\tau|$ errors on the erased coordinates. At least one has at most $\lfloor |\mathcal{E}_\tau|/2 \rfloor$ such errors, and its total Hamming distance from $\mathbf{c}$ is at most $e_\tau + \lfloor |\mathcal{E}_\tau|/2 \rfloor \leq \lfloor (d-1)/2 \rfloor = T_d$. The bounded distance decoder consequently returns $\mathbf{c}$ on that word.

To show that such a threshold exists, choose τ at random with distribution $\Pr[\tau \leq t] = 4t^2$ on $[0, 1/2]$. This random choice is used only in the proof. Coordinate i is then erased with probability $\Pr[\tau \leq t_i] = 4t_i^2$. Put $\rho_i := \text{dist}(y_i, c_i + 2\mathbb{Z})$, the distance from y_i to the parity class specified by the fixed codeword. If $p_i = c_i$, then $\rho_i = t_i$. This coordinate contributes zero to $2e_\tau + |\mathcal{E}_\tau|$ when retained and one when erased, so its expected contribution is $4t_i^2 = 4\rho_i^2$. If $p_i \neq c_i$, then $\rho_i = 1 - t_i$. This coordinate contributes two when retained and one when erased, so its expected contribution is $2(1 - 4t_i^2) + 4t_i^2 = 2 - 4t_i^2$. Moreover, $2 - 4t_i^2 \leq 4(1 - t_i)^2 = 4\rho_i^2$, since $4(1 - t_i)^2 - (2 - 4t_i^2) = 8(t_i - 1/2)^2 \geq 0$. Thus, in either case, the expected contribution is at most $4\rho_i^2$. By linearity of expectation, $\mathbb{E}[2e_\tau + |\mathcal{E}_\tau|] \leq 4 \sum_{i=1}^N \rho_i^2 = 4d_{2,\text{tor}}(\mathbf{y}, \mathbf{c})^2 < d$.

For correctness, within each interval, the erased set and the error count are constant. The expectation bound therefore implies that at least one of the intervals has $2e_\tau + |\mathcal{E}_\tau| < d$. The algorithm tests its midpoint, and one of the two corresponding decoder calls recovers $\mathbf{c}$. This codeword passes the final distance test. Every accepted output belongs to D and satisfies the required distance bound. Distinct codewords have Euclidean distance modulo two at least $\sqrt{d}$, so the triangle inequality permits at most one such codeword. If none exists, every candidate fails the final test, and the algorithm returns $\perp$.

For running time, there are at most $N+1$ threshold intervals and two decoder calls per interval. The thresholds are rational and sorting, checking the outputs take time polynomial in N and $\text{bitlen}(\mathbf{y})$. This proves the claimed complexity bound.

4 Lattice Construction and List Decoding

4.1 The nested tower and its integer lift

We obtain a nested chain by taking intersections of the component codes. To control the determinant of the resulting lattice, we lift a binary basis adapted to this chain to a unimodular integer matrix. We first establish the required lifting lemma.

Lemma 4.1 (Unimodular lift). *For every $\bar{\mathbf{U}} \in \text{GL}_n(\mathbb{F}_2)$, one can deterministically construct in polynomial time a matrix $\mathbf{U} \in \text{GL}_n(\mathbb{Z})$ such that $\mathbf{U} \bmod 2 = \bar{\mathbf{U}}$. A direct implementation produces entries of bit length $O(n^2)$.*

Proof. Gaussian elimination over $\mathbb{F}_2$ produces $m = O(n^2)$ elementary matrices $\bar{\mathbf{E}}_1, \dots, \bar{\mathbf{E}}_m$ such that $\bar{\mathbf{E}}_m \cdots \bar{\mathbf{E}}_1 \bar{\mathbf{U}} = \mathbf{I}_n$. Each $\bar{\mathbf{E}}_t$ either exchanges two rows or has the form $\mathbf{I}_n + \mathbf{e}_i \mathbf{e}_j^\top$ for $i \neq j$, where $\mathbf{e}_i$ denotes the i th standard basis vector; the latter operation adds row j to row i .

Lift each $\bar{\mathbf{E}}_t$ to the corresponding integer elementary matrix $\mathbf{E}_t$, and set $\mathbf{U} = \mathbf{E}_1^{-1} \cdots \mathbf{E}_m^{-1}$. Each factor is unimodular, and reduction modulo two gives $\mathbf{U} \bmod 2 = \bar{\mathbf{U}}$. Starting from $\mathbf{I}_n$, multiplication by each $\mathbf{E}_t^{-1}$ at most doubles the largest absolute entry. Thus the entries of $\mathbf{U}$ have bit length $O(n^2)$, and the construction takes polynomial time.

Fix $a \geq 1$. Suppose that, for $0 \leq j < a$, a binary linear code $D_j \subseteq \mathbb{F}_2^n$ has distance at least d_j and codimension r_j , and set $d_a := 1$. Define $C_j := \bigcap_{t=j}^{a-1} D_t$ with $C_0 \subseteq C_1 \subseteq \cdots \subseteq C_{a-1}$. Write $k_j = \dim(C_j)$. Choose an adapted binary basis $\bar{\mathbf{b}}_1, \dots, \bar{\mathbf{b}}_n$ such that $C_j = \text{span}_{\mathbb{F}_2}\{\bar{\mathbf{b}}_1, \dots, \bar{\mathbf{b}}_{k_j}\}$. We make this choice deterministically. Starting with the empty list, process $C_0, \dots, C_{a-1}$ in order. At each level, scan the ordered reduced-row-echelon basis of C_j ¹ and append each vector that is linearly independent of those already chosen. Finally, scan the standard basis in coordinate order to extend the list to a basis of $\mathbb{F}_2^n$. All steps use polynomial-time linear algebra.

Let $\bar{\mathbf{U}}$ be the matrix with columns $\bar{\mathbf{b}}_1, \dots, \bar{\mathbf{b}}_n$. Apply Lemma 4.1 to obtain $\mathbf{U} \in \text{GL}_n(\mathbb{Z})$, and write $\mathbf{b}_i$ for its i th column, so that $\mathbf{b}_i \bmod 2 = \bar{\mathbf{b}}_i$. With $k_{-1} = 0$, assign $\ell_i = \begin{cases} j, & k_{j-1} < i \leq k_j, \\ a, & i > k_{a-1}. \end{cases}$, then define $\mathcal{L} := \mathbf{U} \text{diag}(2^{\ell_1}, \dots, 2^{\ell_n}) \mathbb{Z}^n$.

Theorem 4.2. *Define $\mathcal{L}$ as above, it satisfies*

$$\lambda_1(\mathcal{L})^2 \geq \min_{0 \leq j \leq a} 4^j d_j, \quad \log_2 \det(\mathcal{L}) \leq \sum_{j=0}^{a-1} (j+1) r_j.$$

An integer basis of $\mathcal{L}$ can be constructed deterministically in time polynomial in n , a , and the total encoding length of the component-code data. Its entries have bit length $O(n^2 + a)$.

Proof. Consider a nonzero vector $\mathbf{x} = \sum_{i=1}^n 2^{\ell_i} z_i \mathbf{b}_i \in \mathcal{L}$, where $z_i \in \mathbb{Z}$, and set $q := \min_i (\ell_i + \nu_2(z_i))$.

If $q < a$, every coefficient $2^{\ell_i} z_i$ is divisible by 2^q , so $\mathbf{y} = \mathbf{x}/2^q \in \mathbb{Z}^n$. The coefficient $2^{\ell_i - q} z_i$ is odd precisely when $\ell_i + \nu_2(z_i) = q$. Thus $\mathbf{y} \bmod 2$ is the sum of the vectors $\bar{\mathbf{b}}_i$ over these indices. Each such index satisfies $\ell_i \leq q$, so $\mathbf{y} \bmod 2 \in C_q \subseteq D_q$. The sum is nonempty, and the vectors $\bar{\mathbf{b}}_i$ are linearly independent; hence $\mathbf{y} \bmod 2 \neq \mathbf{0}$. Therefore $\|\mathbf{x}\|_2^2 = 4^q \|\mathbf{y}\|_2^2 \geq 4^q \text{wt}(\mathbf{y} \bmod 2) \geq 4^q d_q$. If $q \geq a$, every coefficient $2^{\ell_i} z_i$ is divisible by 2^a . Since the vectors $\mathbf{b}_i$ are integral, $\mathbf{x} \in 2^a \mathbb{Z}^n$. As $\mathbf{x} \neq \mathbf{0}$, this gives $\|\mathbf{x}\|_2^2 \geq 4^a$. Together, the two cases prove the $\lambda_1(\mathcal{L})$ bound.

¹ Take any matrix whose columns span C_j and compute the reduced row-echelon form of its transpose. Read the nonzero rows from top to bottom and transpose each into a column vector. The resulting ordered list depends only on C_j .

For the determinant, the matrix $\mathbf{U} \operatorname{diag}(2^{\ell_1}, \dots, 2^{\ell_n})$ is an integer basis of $\mathcal{L}$. Since $\mathbf{U}$ is unimodular,² we have $\det(\mathcal{L}) = 2^{\sum_i \ell_i}$. For every $0 \leq j < a$, exactly $n - k_j$ indices satisfy $\ell_i > j$. Counting each index once for every level below ℓ_i gives $\sum_i \ell_i = \sum_{j=0}^{a-1} (n - k_j)$. Moreover, Lemma 2.2 gives $n - k_j = \operatorname{codim}(C_j) \leq \sum_{t=j}^{a-1} r_t$. Consequently, $\log_2 \det(\mathcal{L}) \leq \sum_{j=0}^{a-1} \sum_{t=j}^{a-1} r_t = \sum_{t=0}^{a-1} (t+1)r_t$.

The intersections and adapted binary basis are computed by polynomial-time linear algebra. By Lemma 4.1, the integer lift $\mathbf{U}$ is computable in polynomial time and has entries of bit length $O(n^2)$. Since $0 \leq \ell_i \leq a$, multiplying its i th column by 2^{ℓ_i} gives basis entries of bit length $O(n^2 + a)$. This proves the algorithmic claims.

4.2 The lattice family in every dimension

For each $n \geq 1$, set

$$a(n) := \max\{0, \lfloor \log_4(n/2) \rfloor\}, \quad d_j := \lceil n/4^j \rceil \quad (0 \leq j < a(n)). \quad (3)$$

We write $a = a(n)$ when the dimension is clear. Take $D_j = D(n, d_j)$ from Theorem 3.4, write $r_j = \operatorname{codim}(D_j)$, and apply the preceding construction to obtain $\mathcal{L}_n$. When $a = 0$, use the empty tower and set $\mathcal{L}_n = \mathbb{Z}^n$. Let A be the absolute constant in Theorem 3.4 and set $C_{\det} := 2^{20\sqrt{2}A}$.

Proposition 4.3. *For every $n \geq 1$, the lattice $\mathcal{L}_n$ is deterministically constructible in $\operatorname{poly}(n)$ time and satisfies*

$$\lambda_1(\mathcal{L}_n) = 2^{a(n)}, \quad \det(\mathcal{L}_n)^{1/n} \leq C_{\det}, \quad \gamma(\mathcal{L}_n) > \frac{n}{8C_{\det}^2}.$$

The algorithm outputs an integer basis of $\mathcal{L}_n$ together with all component-code and lift data, with total encoding length polynomial in n .

Proof. When $a(n) = 0$, the minimum-distance and determinant bounds follow immediately from $\mathcal{L}_n = \mathbb{Z}^n$. Assume now that $a > 0$. For every $0 \leq j < a$, we have $4^j d_j \geq n \geq 4^a$, so Theorem 4.2 gives $\lambda_1(\mathcal{L}_n) \geq 2^a$. Conversely, $\ell_i \leq a$ for every i , and unimodularity gives $\mathbf{U}\mathbb{Z}^n = \mathbb{Z}^n$. Hence

$$2^a \mathbb{Z}^n = \mathbf{U}(2^a \mathbb{Z}^n) \subseteq \mathbf{U} \operatorname{diag}(2^{\ell_1}, \dots, 2^{\ell_n}) \mathbb{Z}^n = \mathcal{L}_n.$$

In particular, $2^a \mathbf{e}_1 \in \mathcal{L}_n$, so $\lambda_1(\mathcal{L}_n) \leq 2^a$, proving equality.

For the determinant, the definition of d_j gives $d_j/n \leq 2 \cdot 4^{-j}$ and $\log_2(2n/d_j) \leq 2j + 1$. Combining these estimates with $r_j \leq A\sqrt{nd_j} \log_2(2n/d_j)$ from Theorem 3.4 yields $r_j/n \leq A\sqrt{2} 2^{-j}(2j + 1)$. The determinant bound in Theorem 4.2 therefore gives

$$\frac{1}{n} \log_2 \det(\mathcal{L}_n) \leq A\sqrt{2} \sum_{0 \leq j < a} \frac{(j+1)(2j+1)}{2^j} \leq 20\sqrt{2}A.$$

² An arbitrary 0–1 lift of a binary basis need only have odd determinant; its absolute determinant can be large. The unimodular lift removes this uncontrolled contribution.

Here $\sum_{j=0}^{\infty} (j+1)(2j+1)2^{-j} = 20$, as follows by differentiating the geometric series. This proves the claimed bound.

For every $n \geq 1$, including the case $a = 0$, (3) gives $4^a > n/8$. Consequently, $\gamma(\mathcal{L}_n) = \lambda_1(\mathcal{L}_n)^2 / \det(\mathcal{L}_n)^{2/n} \geq 4^a / C_{\det}^2 > n / (8C_{\det}^2)$.

Finally, there are $a = O(\log n)$ levels, and Theorem 3.4 constructs each component code and its decoding data in $\text{poly}(n)$ time. Applying Theorem 4.2 then gives the stated construction time and polynomial total encoding length.

4.3 List decoding

Define the auxiliary lattices $\mathcal{M}_j := \mathbf{U} \text{diag}(2^{\max\{\ell_1-j, 0\}}, \dots, 2^{\max\{\ell_n-j, 0\}}) \mathbb{Z}^n$ for $0 \leq j \leq a$, and set $C_a = \mathbb{F}_2^n$. At level j , the decoder recovers the parity vector of the unknown point in $\mathcal{M}_j$, subtracts an integer lift of that vector, and divides by two to obtain a decoding problem in $\mathcal{M}_{j+1}$. For the empty tower, take $\mathbf{U} = \mathbf{I}_n$ and $\ell_i = 0$. The following identities justify this recursion.

Lemma 4.4. *One has $\mathcal{M}_0 = \mathcal{L}_n$, $\mathcal{M}_a = \mathbb{Z}^n$, and $\mathcal{M}_j \bmod 2 = C_j$. For every $0 \leq j < a$,*

$$\{\mathbf{x} \in \mathcal{M}_j : \mathbf{x} \bmod 2 = \mathbf{0}\} = 2\mathcal{M}_{j+1}. \quad (4)$$

Proof. The identity $\mathcal{M}_0 = \mathcal{L}_n$ follows from the definition, while $\ell_i \leq a$ and unimodularity give $\mathcal{M}_a = \mathbf{U}\mathbb{Z}^n = \mathbb{Z}^n$. Modulo two, precisely the columns with $\ell_i \leq j$ can contribute, so $\mathcal{M}_j \bmod 2 = \text{span}_{\mathbb{F}_2}\{\bar{\mathbf{b}}_i : \ell_i \leq j\} = C_j$.

For the kernel identity, write $\mathbf{x} = \mathbf{U}\mathbf{t}$. Since $\mathbf{U}$ and $\mathbf{U}^{-1}$ are integral, $\mathbf{x} \in 2\mathbb{Z}^n$ if and only if $\mathbf{t} \in 2\mathbb{Z}^n$. Thus $\mathbf{x} \in \mathcal{M}_j \cap 2\mathbb{Z}^n$ if and only if $2^{\max\{\ell_i-j, 1\}}$ divides t_i for every i . The identity $\max\{\ell_i-j, 1\} = 1 + \max\{\ell_i-(j+1), 0\}$ shows that this is equivalent to $\mathbf{x} \in 2\mathcal{M}_{j+1}$, proving (4).

The recursion terminates in $\mathbb{Z}^n$, where the required list can be enumerated directly.

Lemma 4.5 (List decoding in $\mathbb{Z}^n$). *For every $\mathbf{u} \in \mathbb{Q}^n$, the set $\mathcal{Z}(\mathbf{u}) := \{\mathbf{z} \in \mathbb{Z}^n : \|\mathbf{z} - \mathbf{u}\|_2^2 < 1/2\}$ has at most two elements and can be computed with $O(n)$ exact arithmetic tests and polynomial bit complexity.*

Proof. Let $\mathbf{q}$ be a coordinatewise nearest integer vector to $\mathbf{u}$, with ties broken deterministically, and put $\mathbf{v} = \mathbf{u} - \mathbf{q}$ and $b = \|\mathbf{v}\|_2^2$. Since $\mathbf{q}$ is a nearest point in $\mathbb{Z}^n$, the list is empty if $b \geq 1/2$.

Otherwise, include $\mathbf{q}$. Any other admissible point differs from $\mathbf{q}$ in exactly one coordinate: each changed coordinate contributes at least $1/4$ to its squared distance from $\mathbf{u}$. Moreover, the change must be one unit toward u_i . Thus it suffices to test $\mathbf{q} + \text{sgn}(v_i)\mathbf{e}_i$ for each i with $v_i \neq 0$, accepting it precisely when $b + 1 - 2|v_i| < 1/2$.

Distinct candidates other than $\mathbf{q}$ have distance $\sqrt{2}$, so at most one can lie in the open ball of radius $1/\sqrt{2}$. The list therefore has at most two elements. Computing b once permits each candidate test to use a constant number of arithmetic operations, and only the accepted vectors need to be written in full.

The list-decoding algorithm uses the public construction data and the GMD decoder of [Lemma 3.6](#).

1. Set $\mathbf{y}_0 = \mathbf{y}$.
2. For $j = 0, \dots, a-1$, apply [Lemma 3.6](#) to $\mathbf{y}_j$ and the supercode D_j . If decoding fails or the returned codeword does not belong to C_j , return the empty list. Otherwise, denote the returned parity vector by $\mathbf{c}_j$ and express $\mathbf{c}_j = \sum_{i \leq k_j} \alpha_{j,i} \mathbf{b}_i$, where $\alpha_{j,i} \in \{0, 1\}$. Set $\mathbf{v}_j = \sum_{i \leq k_j} \alpha_{j,i} \mathbf{b}_i \in \mathcal{M}_j$ and $\mathbf{y}_{j+1} = (\mathbf{y}_j - \mathbf{v}_j)/2$.
3. Enumerate $\mathcal{Z}(\mathbf{y}_a)$ using [Lemma 4.5](#). For each $\mathbf{z}$ in this list, output $\mathbf{x} = \sum_{j=0}^{a-1} 2^j \mathbf{v}_j + 2^a \mathbf{z}$.

Theorem 4.6 (List decoding). *For every $n \geq 1$ and $\mathbf{y} \in \mathbb{Q}^n$, the algorithm returns exactly*

$$\text{List}\left(\mathcal{L}_n, \mathbf{y}, \frac{\lambda_1(\mathcal{L}_n)}{\sqrt{2}}\right).$$

The list has at most two elements, and the running time is $\text{poly}(n, \text{bitlen}(\mathbf{y}))$.

Proof. For $a = 0$, the claim is [Lemma 4.5](#). Assume $a > 0$ and put $R = 2^a/\sqrt{2}$. Since $2 \cdot 4^a \leq n \leq 4^j d_j$, we have $R/2^j \leq \sqrt{d_j}/2$ for every $0 \leq j < a$.

Suppose $\mathbf{x} \in \mathcal{L}_n$ satisfies $\|\mathbf{y} - \mathbf{x}\|_2 < R$, and write $\mathbf{e} = \mathbf{y} - \mathbf{x}$. We prove inductively that decoding succeeds at every level and $\mathbf{y}_j = \mathbf{x}_j + \mathbf{e}/2^j$ for some $\mathbf{x}_j \in \mathcal{M}_j$. Initially, take $\mathbf{x}_0 = \mathbf{x}$. At level j , the parity vector $\mathbf{x}_j \bmod 2$ belongs to $C_j \subseteq D_j$, and its Euclidean distance modulo two from $\mathbf{y}_j$ is at most $\|\mathbf{e}\|_2/2^j < \sqrt{d_j}/2$. Thus [Lemma 3.6](#) returns this vector, which passes the C_j membership test. Since $\mathbf{v}_j$ has the same parity, (4) gives $\mathbf{x}_{j+1} = (\mathbf{x}_j - \mathbf{v}_j)/2 \in \mathcal{M}_{j+1}$, and the error identity follows. In particular, a decoding failure implies that the desired list is empty.

When all levels succeed, put $\mathbf{w} = \sum_{j=0}^{a-1} 2^j \mathbf{v}_j$. Then $\mathbf{y}_a = (\mathbf{y} - \mathbf{w})/2^a$, and the preceding induction shows that every nearby lattice point has the form $\mathbf{x} = \mathbf{w} + 2^a \mathbf{z}$ for some $\mathbf{z} \in \mathbb{Z}^n$. Conversely, every such vector belongs to $\mathcal{L}_n$: starting from $\mathbf{x}_a = \mathbf{z}$, the reverse recursion $\mathbf{x}_j = \mathbf{v}_j + 2\mathbf{x}_{j+1}$ stays in $\mathcal{M}_j$ by (4). Finally, $\|\mathbf{y} - \mathbf{x}\|_2 = 2^a \|\mathbf{y}_a - \mathbf{z}\|_2$, so the desired lattice list corresponds exactly to $\mathcal{Z}(\mathbf{y}_a)$. This proves both correctness and the bound of two on the list size.

All construction data are computable from n in $\text{poly}(n)$ time. Each of the $a = O(\log n)$ levels uses $O(n)$ calls to a polynomial-time component decoder, together with membership tests and linear algebra. The lift vectors have polynomial bit length, and $\mathbf{y}_j = 2^{-j} \left(\mathbf{y} - \sum_{t < j} 2^t \mathbf{v}_t \right)$ shows that numerator and denominator lengths remain polynomial in n and $\text{bitlen}(\mathbf{y})$. All distance tests compare rational squared distances with rational thresholds. Together with [Lemma 4.5](#), this proves the running-time bound.

Remark 4.7 (Larger radii). For $a > 0$, the decoder extends to the open radius $R_n := \sqrt{n}/2$, since $R_n/2^j \leq \sqrt{d_j}/2$ at every coded level. The terminal radius is $\rho_n := R_n/2^a \in [1/\sqrt{2}, \sqrt{2})$. Every terminal candidate differs from a coordinate-wise nearest integer vector in at most seven coordinates, by at most one in each.

Testing these $O(n^7)$ candidates gives polynomial-time enumeration. The relative radius $R_n/\lambda_1(\mathcal{L}_n) = \rho_n$ still has minimum $1/\sqrt{2}$, attained when $n = 2 \cdot 4^a$.

4.4 Normalization

Define

$$s_n := \min\{s \in \mathbb{Z}_{\geq 1} : s^2 4^a \geq n\} = \lceil \sqrt{n}/2^a \rceil, \quad \widehat{\mathcal{L}}_n := s_n \mathcal{L}_n.$$

Since $n/8 < 4^a \leq n$, we have $1 \leq s_n \leq 3$. By [Proposition 4.3](#),

$$\sqrt{n} \leq \lambda_1(\widehat{\mathcal{L}}_n) = s_n 2^a < \sqrt{n} + 2^a \leq 2\sqrt{n}, \quad \det(\widehat{\mathcal{L}}_n)^{1/n} \leq 3C_{\det}.$$

An integer basis of $\widehat{\mathcal{L}}_n$ is obtained by multiplying the basis of $\mathcal{L}_n$ by s_n , preserving polynomial construction time and encoding length.

To list decode in $\widehat{\mathcal{L}}_n$, apply [Theorem 4.6](#) to $\mathbf{y}/s_n$ and multiply each returned vector by s_n . This returns precisely the lattice vectors at distance less than $\lambda_1(\widehat{\mathcal{L}}_n)/\sqrt{2}$ from $\mathbf{y}$. The list still has at most two elements, and the running time remains $\text{poly}(n, \text{bitlen}(\mathbf{y}))$.

Proof (Proof of [Theorem 1.1](#)). Take $C = 3C_{\det}$. The above normalization gives the claimed bounds. The construction guarantees follow from [Proposition 4.3](#) and $s_n \leq 3$. The scaled list decoder gives the decoding guarantees.

Use of Generative AI

During the preparation of this manuscript, the authors used OpenAI Codex for assistance with exposition, literature review, and exploratory discussions of proof details. The authors take full responsibility for the content of the final manuscript, including the correctness of all mathematical claims, proofs, and references.

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A Bounded-distance decoding of shortened BCH codes

We give a self-contained description of the classical syndrome decoder for the shortened BCH codes used in the paper. We use Peterson’s linear-algebraic approach [23]; the error-locator polynomial can also be computed by the Berlekamp–Massey algorithm [19].

Let $1 \leq h \leq N$, let $m = \lceil \log_2(N+1) \rceil$, and let $x_1, \dots, x_N$ be distinct nonzero elements of $\mathbb{K} = \mathbb{F}_{2^m}$. Recall that $B(N, h) = \{\mathbf{c} \in \mathbb{F}_2^N : \sum_{j=1}^N c_j x_j^r = 0, \forall r \in [h-1]\}$, $T_h := \lfloor (h-1)/2 \rfloor$. We describe the decoder directly on these N coordinates. Adding zero coordinates at the unused elements of $\mathbb{K}^\times$ gives the same syndromes.

For a received word $\mathbf{w} \in \mathbb{F}_2^N$, the algorithm proceeds as follows.

1. Compute the syndromes $S_r := \sum_{j=1}^N w_j x_j^r$, $r \in [h-1]$. If all syndromes vanish, return $\mathbf{w}$. If $T_h = 0$ and a syndrome is nonzero, return $\perp$. In particular, for $h = 1$ the algorithm always returns $\mathbf{w}$.
2. Form the Hankel matrix $\mathbf{H} := (S_{r+s-1})_{1 \leq r, s \leq T_h}$ and compute $\nu = \text{rank}_{\mathbb{K}}(\mathbf{H})$. If $\nu = 0$, return $\perp$. Let $\mathbf{H}_\nu$ be its leading $\nu \times \nu$ submatrix. If $\mathbf{H}_\nu$ is singular, return $\perp$. Otherwise solve $\mathbf{H}_\nu \begin{pmatrix} \lambda_\nu \\ \lambda_{\nu-1} \\ \vdots \\ \lambda_1 \end{pmatrix} = - \begin{pmatrix} S_{\nu+1} \\ S_{\nu+2} \\ \vdots \\ S_{2\nu} \end{pmatrix}$ over $\mathbb{K}$, and set $A(X) := 1 + \lambda_1 X + \dots + \lambda_\nu X^\nu$.
3. Evaluate $A(x_j^{-1})$ for every $j \in [N]$ and put $J := \{j : A(x_j^{-1}) = 0\}$. If $|J| \neq \nu$, return $\perp$. Otherwise form $\mathbf{c} \in \mathbb{F}_2^N$ by setting $c_j = w_j + 1$ for $j \in J$ and $c_j = w_j$ elsewhere. Return $\mathbf{c}$ if $\mathbf{c} \in B(N, h)$ and $\text{wt}(\mathbf{w} - \mathbf{c}) \leq T_h$; otherwise return $\perp$.

All syndrome indices used by the algorithm are at most $2T_h \leq h-1$. Membership in the last step is checked using all $h-1$ parity checks defining $B(N, h)$.

Proposition A.1. *For every $\mathbf{w} \in \mathbb{F}_2^N$, the algorithm returns the unique $\mathbf{c} \in B(N, h)$ satisfying $\text{wt}(\mathbf{w} - \mathbf{c}) \leq T_h$, if such a codeword exists, and returns $\perp$ otherwise. It uses $O(Nh + T_h^3)$ operations in $\mathbb{K}$ and has deterministic polynomial bit complexity in N .*

Proof. Since $d(B(N, h)) \geq h > 2T_h$, there is at most one codeword within distance T_h of $\mathbf{w}$. Every returned word belongs to $B(N, h)$ and satisfies the distance bound, by the explicit tests. It remains to prove that the algorithm finds such a codeword whenever one exists.

Suppose $\mathbf{c} \in B(N, h)$ and $t := \text{wt}(\mathbf{w} - \mathbf{c}) \leq T_h$. If $t = 0$, all syndromes vanish and the first step returns $\mathbf{w} = \mathbf{c}$. Assume $t > 0$, and write $i_1, \dots, i_t$ for the error positions and $z_b := x_{i_b}$ for $1 \leq b \leq t$. The code is binary, so $S_r = \sum_{b=1}^t z_b^r$, $r \in [h-1]$. Let $\mathbf{V} = (z_b^{r-1})_{1 \leq r \leq T_h, 1 \leq b \leq t}$. Then $\mathbf{H} = \mathbf{V} \text{diag}(z_1, \dots, z_t) \mathbf{V}^\top$. The z_b are distinct and nonzero, so $\mathbf{V}$ has column rank t . Its leading $t \times t$ submatrix $\mathbf{V}_t$ is an invertible Vandermonde matrix. Hence $\mathbf{H}_t = \mathbf{V}_t \text{diag}(z_1, \dots, z_t) \mathbf{V}_t^\top$ is invertible. Together with the factorization of $\mathbf{H}$, this gives $\text{rank}(\mathbf{H}) = t$. In particular, the syndromes do not all vanish, the algorithm obtains $\nu = t$, and $\mathbf{H}_\nu$ is invertible.

Consider the error-locator polynomial $A^*(X) := \prod_{b=1}^t (1 - z_b X) = 1 + \lambda_1^* X + \dots + \lambda_t^* X^t$. For $1 \leq r \leq t$, $S_{r+t} + \sum_{k=1}^t \lambda_k^* S_{r+t-k} = \sum_{b=1}^t z_b^{r+t} A^*(z_b^{-1}) = 0$. Thus $(\lambda_t^*, \dots, \lambda_1^*)^\top$ solves the system in the second step. Its solution is unique, so

$\Lambda = \Lambda^*$. Consequently, $\Lambda(x_j^{-1}) = 0 \iff j \in \{i_1, \dots, i_t\}$. The third step therefore flips exactly the erroneous bits and returns $\mathbf{c}$.

Computing all syndromes takes $O(Nh)$ field operations. Gaussian elimination for the rank computation and the linear system takes $O(T_h^3)$ operations. Computing the inverses and evaluating Λ takes $O(N(T_h+1))$ operations, and the final verification takes $O(Nh)$ operations. Each field element has $m = O(\log(N+1))$ bits, and field operations have $\text{poly}(m)$ bit complexity. This proves the claimed running time.